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Use of a Novel Standardized Administration Protocol Reduces Agitation Pro Re Nata (PRN) Medication Requirements: The Birmingham Agitation Management (BAM) Initiative

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Abstract

Background: Agitation management is a principal challenge on inpatient psychiatric units. Overreliance on common prescribing strategies of pro re nata (PRN) medication administration is problematic, given the tendencies to have overlapping or unclear indications. **Objective:** Piloted project to determine whether a standardized protocol for agitation intervention may reduce PRN medication administration. **Methods:** The Birmingham Agitation Management (BAM) interdisciplinary team uniquely connected the Brøset Violence Checklist (BVC) for assessment of agitation severity to a standardized PRN medication order set. Nurses on the piloted unit were trained on how to score the BVC and administer medications. Patients were assessed by the BVC every 4 hours and, based on their score, would receive no medication, low-dose benzodiazepine, high-dose benzodiazepine, or high-dose benzodiazepine plus antipsychotic. The primary end point compared the number of PRNs administered after novel protocol implementation with a retrospective cohort. Secondary measures included analysis of medication-related effects, seclusion, and physical restraint rates. **Results:** 377 patients were included in the final analyses (184 pre-BAM, 193 BAM intervention group). No significant differences were seen in patient characteristics between groups. The total number of PRNs administered decreased by 42.5%, with both the mean and median number of administrations decreasing significantly (95% confidence interval [CI] = [1.68-5.75]; $P < 0.001$). A trend was noted between the number of PRNs administered and seclusion rates, but did not reach statistical significance (95% CI = [–7.28 to 60.31]; $P = 0.124$). **Conclusions:** In seemingly the first initiative of its kind, we found that a standardized agitation management protocol can help decrease the total number of PRN administrations for agitation without worsening of restraint rates and may possibly reduce the risk of adverse effects. These results require validation in specific, larger populations.

Keywords

scoring system, agitation, pro re nata administration, inpatient psychiatry, Birmingham Agitation Management

Introduction

Management of agitation and aggression remains an important part of psychiatric care. The prevalence of agitation in patients with schizophrenia or mood disorder is approximately 11% to 13%, with higher rates among patients with alcohol use disorders (25%) or substance misuse (35%).[1] Up to 10% of emergency psychiatric interventions involve the management of agitated behaviors.[2,3] In acute inpatient psychiatric units, agitation disrupts patient interactions during patient rounds or educational groups, and places other patients and staff at risk of harm. Aggression, a symptom of agitation, may present as verbal threats, physical threats, or self-injury. The pathophysiology of agitation and aggression is complex, and multiple mechanisms have been proposed. As such, the management of agitation varies patient by patient, depending on the scenario.

The main goal of agitation management is to calm the patient using the least restrictive manner, including verbal de-escalation or removing the patient from a specific environment, prior to use of physical restraints or administration of medications with the goal of de-escalation or sedation.[4] Assessment of the underlying etiology of agitation is critical for effective treatment.[5] The intervention selected should be proportionate to the severity of the risk posed by the acute disturbance.[6] Agitation exists on a spectrum, which may be considered mild, moderate, or severe depending on specific symptoms exhibited.[7] It is unlikely that a mildly agitated patient requires physical restraint, or that a severely agitated patient will settle down with only verbal de-escalation, hence early recognition of agitation is important in reduction of high-risk situations created by need for restraints and involuntary sedation.[8] Furthermore, addressing agitation early may help prevent escalation to assault or the requirement of physical restraints or seclusion.

Despite the pervasive nature of the problem, existing literature guidelines on agitation treatment approaches are inconsistent with respect to which medication is optimal, and practices vary widely by country, region, and institution.[3] Antipsychotics and benzodiazepines are considered mainstays of therapy for agitation. Haloperidol has been the most widely utilized antipsychotic medication for agitation.[9-11] While second-generation antipsychotics are reportedly effective and well-tolerated, evidence is limited when comparing these with first-generation agents.[11] One conventional regimen for agitation management is the "5-2-1," which includes 5 mg haloperidol, 2 mg lorazepam, and 1 mg benztropine, or the "B-52" regimen, which replaces benztropine with 50 mg diphenhydramine.[12] These medication combinations, while possibly effective, may be no more effective than use of any of the components individually.[13] For safety purposes, minimizing polypharmacy is optimal. In addition, administering excessive medication may elicit intolerable adverse effects or result in prolonged oversedation, prohibiting patients from engaging in therapeutic activities.[14] Overmedication is especially concerning in older adult patients, who are more susceptible to sedation.[15]

Guidelines vary with respect to the recommended first-line agent(s); however, barring contraindications, benzodiazepines may be utilized alone for undifferentiated agitation.[10,14] The National Institute for Health and Care Excellence (NICE) recommend benzodiazepines prior to attempting haloperidol plus promethazine. First-generation antipsychotics (FGAs) are also recommended for undifferentiated psychosis, but use alone is strongly discouraged due to the risk of extrapyramidal adverse effects.[10,16] For those with agitation secondary to known psychiatric disorders, the Project BETA psychopharmacology workgroup recommends use of oral risperidone or olanzapine plus or minus benzodiazepines as first-line options, whereas haloperidol with benzodiazepines are recommended second-line.[14] Finally, the British Association for Psychopharmacology and the National Association of Psychiatric Intensive Care and Low Secure Units consensus guidelines recommend an oral antipsychotic or lorazepam for prerapid tranquilization (RT), monotherapy with an intramuscular (IM) lorazepam or IM antipsychotics for RT, or combination IM lorazepam plus haloperidol for RT when oral administration is not feasible.[6]

Although regulatory agencies such as the Joint Commission and Center for Medicare and Medicaid emphasize the need for reduction in the use of seclusions and restraints in psychiatric inpatients, there is an absence of quality guidelines for psychopharmacological management of agitation.[17,18] Relatedly, numerous studies have evaluated PRN administrations for psychotropic medications in the psychiatric population. PRN orders can lack specific instruction on when to administer, creating nonstandardized levels of medical decision-making for nursing, with resulting secondary safety risks, including administering beyond maximum daily doses of antipsychotics, PRN overuse or without rationale, or PRN administration near scheduled medication dosing times with resulting oversedation.[19-26] Furthermore, giving PRN antipsychotics on top of scheduled antipsychotics may lead to a higher incidence of adverse effects and confuse the clinical picture with respect to which antipsychotic a patient may be responding.[23]

Measures to promote standardization of pharmacological management in individuals with psychiatric disorders are sparsely implemented or reported.[27] However, standardization using an order set built into the electronic health record may be an integral part of improving patient care.[28] Systematic monitoring for agitation utilizing an accepted

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scale (including the Brøset Violence Checklist [BVC], positive and negative syndrome scale–excited component [PANSS-EC], or the Clinical Global Impression [CGI]) may be the first step in overall management.[4] Routine nursing monitoring for development of serious agitation events is analogous to telemetry monitoring on a cardiac unit.[8] Such an alerting protocol not only can help preemptively reduce PRN medication safety-related issues but also alert members of the health care team to determine the best multimodal course of action for a given patient.

Description of Problem and Creation of the Birmingham Agitation Management (BAM) Protocol

Our team noted and questioned the wide assortment of PRNs ordered for agitation, most with seemingly vague instructions on when to administer them. We postulated that, with standardizing prescribing and order clarity, less-agitated patients would require fewer PRN medications. Furthermore, preemptively administering PRNs to a patient who is only mildly agitated may prevent multiple medications from being given, and may ultimately mitigate progression to severe agitation, requiring seclusion and physical restraints. For those severely agitated patients, we presumed administering as few PRN agitation medications as necessary was still prudent, to prevent oversedation and adverse drug reactions. To guide the pharmacologic aspects of agitation management and optimize communication of expectations concerning PRN ordering and administration across care providers, this interdisciplinary team embarked on a pilot study entitled "The BAM" project. The primary purpose of our project was to devise a standardized approach to psychopharmacological management of agitation in efforts to reduce the total number of PRNs administered, while secondarily determining any medication-related effects and differences with rates of seclusion and physical restraints. Our effort is seemingly the first initiative to tie a rating scale for agitation to a distinct PRN order set for management.

Methods

This observational improvement initiative was piloted from June 2017 to October 2020 in a high acuity, inpatient psychiatric unit comprising 20 beds within the Center of Psychiatric Medicine at the University of Alabama at Birmingham Medical Center (UABMC), a key safety net hospital in the state accepting all patients regardless of health insurance status. On average, 55% of patients in this unit are involuntarily admitted and many have been previously committed to state hospitals or are considered treatment resistant. The primary outcome measure was the mean and median number of PRN agitation medications administered to cohorts of patients treated before implementation of the BAM protocol (ie, pre-BAM) versus after implementation (ie, BAM) for all possible medications. Secondary end points included the mean and median number of agitation medications administered before and after implementation of BAM for only those medications utilized in the BAM protocol (ie, haloperidol, lorazepam, ziprasidone, and diazepam), as well as the difference in minutes spent in seclusion and physical restraints preimplementation and postimplementation. An internal UABMC institutional review board (IRB) expedited review approval was obtained under program evaluation and quality improvement (IRB-300008287). No participant consent was required due to the quality improvement status, aimed at enhancing health care outcomes. The initiative was led by 2 board-certified psychiatrists and a psychiatric pharmacist who served as the clinician champions, as well as an interdisciplinary team of nursing leaders, nurse educators, hospital administrators, and hospital informatics personnel. This team met monthly over 3 years to ensure the smooth implementation of the initiative.

All patients admitted to the piloted unit from July 2019 to October 2020, who were between the ages of 18 and 64 years, were evaluated for appropriateness for inclusion. Those admitted with a diagnosis of Schizophrenia, Schizoaffective disorder, Psychosis not otherwise specified (NOS), Bipolar affective disorder (mania, mixed, depression with psychotic symptoms), Substance-induced psychotic/mood disorder, Impulse control disorder and other comorbidities were included in the analyses. Exclusions to entry included pregnancy, increased risk of falls, delirium, no baseline electrocardiogram, basic metabolic panel or complete blood count on file, sleep apnea, or those requiring the evaluation through the Clinical Institute Withdrawal Assessment for Alcohol (CIWA) protocol for alcohol or benzodiazepine withdrawals. The comparator group (ie, pre-BAM) were all patients meeting the inclusion criterion and who were admitted to the same unit from June 2017 to January 2018.

We selected a standardized assessment tool (the BVC; Table 1) to determine agitation severity, which we then tied to corresponding PRN medications orders. The BVC is a validated 6-item checklist, which has shown satisfactory specificity and sensitivity for predicting violence on inpatient psychiatric units, forensic psychiatry units, and in emergency departments.[7,29-31] The BVC reliably predicts an increased risk of violence with every additional point scored.[8] Permission to utilize the BVC scoring system was granted by the creators. Our team chose this assessment tool over other available tools due to the ease of nursing training and subsequent administration. All patients underwent scoring based on the BVC every 4 hours throughout their entire inpatient stay. Nursing leaders and psychiatrists underwent online training to learn how to administer BVC and the nursing leaders/educators subsequently trained frontline

Table 1. Brøset Violence Checklist With Corresponding Operationalization and Scoring.

Confused	Appears obviously confused and disoriented. May be unaware of time, place, or person.	
Irritable	Easily annoyed or angered. Unable to tolerate the presence of others.	
Boisterous	Behavior is overtly "loud" or noisy. For example slams doors, shouts out when talking, and so on.	
Physically threatening	Where there is a definite intent to physically threaten another person. For example, the taking of an aggressive stance; the grabbing of another person's clothing; the raising of an arm, leg, making of a fist, or modeling of a headbutt directed at another.	
Verbally threatening	A verbal outburst that is more than just a raised voice and where there is a definite intent to intimidate or threaten another person. For example, verbal attacks, abuse, name-calling, or verbally neutral comments uttered in a snarling aggressive manner.	
Attacking objects	An attack directed at an object and not an individual. For example, the indiscriminate throwing of an object; banging or smashing windows; kicking, banging, or headbutting an object; or the smashing of furniture.	
Scoring: Components are rated based on *absence* (0) or *presence* (1)		
Total score	0	The risk of violence is small.
	1-2	The risk of violence is moderate; preventive measures should be taken.
	3-6	The risk of violence is high; preventative measures should be taken and plans about how to manage an attack made.

Source: Adapted with permission from the authors of BVC.

inpatient nurses over several weeks in administering BVC on the psychiatric unit. Following training, BVC rating scale and PRN medication orders were inserted in the electronic medical record by informatics and the project was actively implemented on the inpatient unit.

The standard PRN regimen for all participants included lorazepam plus or minus haloperidol (Table 1). No other PRN medications for agitation were ordered outside of the BAM order set. Our decision to utilize benzodiazepines and an FGA (haloperidol) as the first-line option is in line with the historical recommendations for undifferentiated and psychotic agitation management.[13] We opted not to utilize IM olanzapine due to the risk of excess sedation when administered with parenteral benzodiazepines.[32] Doses of medications were adopted from prior studies and based on appropriate oral (PO) to IM conversions.[27,33-35] In a stepwise approach, verbal de-escalation techniques were attempted prior to medication administration. Lower doses of the benzodiazepine were given for a score of 1 on the BVC, higher doses of the benzodiazepine were administered for a score 2 on BVC, and antipsychotics were only given if a patient scored a 3 or more on the BVC. PO administrations were preferred over IM if feasible. Patients determined to be intolerant of either lorazepam or haloperidol were switched to an alternative regimen consisting of diazepam or ziprasidone. In addition, patients who failed to respond to lorazepam plus haloperidol could be switched to diazepam plus haloperidol, lorazepam plus ziprasidone, or diazepam plus ziprasidone. In the instance where a patient failed to respond to the standard regimen and an alternative regimen, they were removed from the BAM protocol and study and placed on various other psychotropic medications for agitation. Failure to respond was dictated by continued agitation or aggression even after use of the max doses of both medications within the chosen regimen.

The total number of PRNs administered to participants included specifically for agitation were counted during the 7-month pre-BAM period and the 15-month intervention period. If a patient received haloperidol plus lorazepam per protocol, then this was counted as 2 medications. If diphenhydramine was written for agitation in the pre-BAM protocol, then this was also included in the total number of PRNs. Scheduled benzodiazepines or antipsychotics and PRNs of any psychotropic prescribed for anxiety or sleep were not included. PRN administration counts were stratified into 4 groups based on the number of doses given (ie, 0, 1-10, 11-20, and >20 administrations). Patients were stratified based on diagnoses for analysis. Those with schizophrenia, schizoaffective disorder, or schizophreniform were segregated into "schizophrenia spectrum disorders."

We utilized descriptive and inferential, parametric, and nonparametric statistics to evaluate the project outcomes. Descriptive data, cross tabulation counts, and percentages were used to evaluate the patient characteristics, including counts associated with pre-BAM versus BAM patient volume, age/age range, sex, race, diagnoses, and PRN medication administration. The (2-sample) t test and the Mann–Whitney U rank-sum test, evaluating both mean and median, were used to compare pre-BAM with BAM total number of PRNs administered and physical restraint and seclusion minutes due to challenges to the assumption of a normal distribution of the variables evaluated.

Results

Demographic Data

A total of 377 patients were included in the primary and secondary analyses (184 in pre-BAM and 193 BAM). Figure 1 depicts the trial profile for screening and inclusion into the

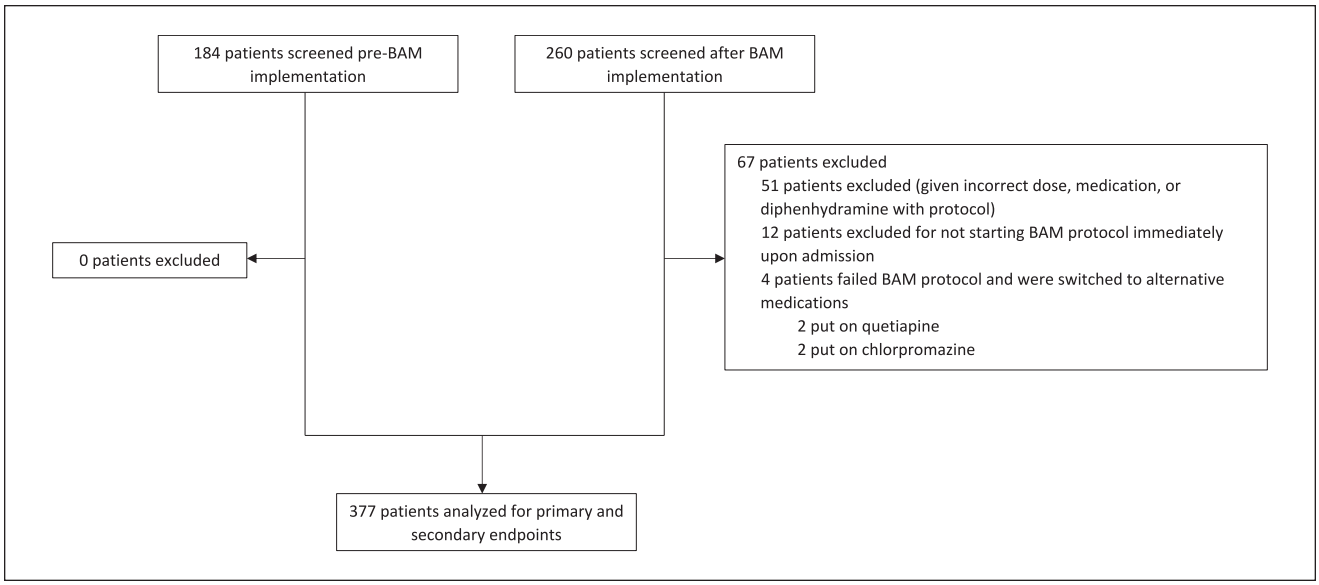

Figure 1. Trial profile for screening and inclusion into data analysis.
Abbreviation: BAM, Birmingham Agitation Management.

analysis. Two-hundred-and-sixty patients were screened postimplementation, but 67 patients were excluded due to administering an incorrect dose, medication, or diphenhydramine with protocol (51); inaccurate protocol start time (12); or failure to respond to the standard BAM regimens (4). No patients were excluded from pre-BAM analysis after screening inclusion. The patient demographics for preimplementation and postimplementation groups can be found in Table 2. No significant differences were apparent when examining the average patient age, length of stay, race, diagnoses, or concomitant medications prescribed. The majority of patients were African American (69% pre-BAM and 73.1% BAM) and male (65.8% pre-BAM and 72% BAM).

Number of Administrations

Fifteen-hundred-and-eight (1508) PRN medications were administered pre-BAM and 867 as-needed medications were administered during BAM (a 42.5% reduction). Forty-two of the 184 patients (22.8%) pre-BAM had no PRN administrations, whereas 77 of the 193 BAM patients (40%) had no PRN administrations. Of the BAM administrations for the standard regimen, lorazepam 2 mg PO, lorazepam 3 mg PO, lorazepam 2 mg IM, and lorazepam 3 mg IM were administered 131, 166, 16, and 199 times, respectively; haloperidol 5 mg PO or IM was administered 107 times and 170 times, respectively. Seven-hundred-and-sixty-six (766) of the 1508 (50.8%) pre-BAM administrations were given PO, compared with 440 (50.7%) BAM.

Twenty-eight patients (20 pre-BAM vs 8 BAM) required >20 medications during their stay. These patients were more likely to be male (78.6%), AA (85.7%), with a longer duration of stay (32.5 days) and have a schizophrenia spectrum disorder diagnosis (60.7%). A total of 67.9% of these patients were on scheduled benzodiazepines or mood stabilizers, respectively, and 100% were on scheduled antipsychotics. These demographics were similar to those not receiving any PRNs during their admission (76.5% male and 77.3% AA). No differences in scheduled antipsychotic or mood stabilizer use were seen between the total population analyzed and those not receiving any PRNs.

Primary and Secondary End points

Table 3 depicts primary and secondary analyses data. For the primary analysis, a statistically significant decrease in the total number of PRNs was found when comparing both the average number of administrations before and after protocol implementation (8.21 ± 11.3 vs 4.39 ± 8.7; 95% confidence interval [CI] = [1.68-5.75]; $P < 0.001$). This significance held true when comparing the median number of administrations as well ($P < 0.001$). Figure 2 displays the average number of PRNs per month before and after protocol implementation. For the secondary analyses, when assessing for medication-related effects, a statistically significant difference was not seen after comparing the average number of only those medications used in the protocol pre-BAM versus BAM (ie, haloperidol, lorazepam, ziprasidone, and diazepam), with a P value of 0.190 (5.57 ± 7 vs 4.49 ± 8.7; 95% CI = [–0.534 to 2.68]). However, when comparing the median values, the difference reached statistical significance ($P = 0.023$). Whereas the average number of minutes spent in seclusion went down from 69.2 to 47.2, the average number of minutes

Table 2. Birmingham Agitation Management Protocol Medication Regimen.

	Medication	Dose	Route	Order comments
Standard regimen	Lorazepam	2 mg	PO IM	For agitation assessment score = 1, oral preferred. Give IM if oral not feasible.
		3 mg	PO IM	For agitation assessment score = 2, give only lorazepam. For agitation assessment score 3 or greater, give with haloperidol.[a] Oral preferred. Give IM if oral not feasible.
	Haloperidol	5 mg	PO IM	For agitation assessment score 3 or greater, give with lorazepam.[c] Oral preferred. Give IM if oral not feasible.
Alternative options[b]	Diazepam	10 mg	PO IM	For agitation assessment score = 1, oral preferred. Give IM if oral not feasible.
		15 mg	PO IM	For agitation assessment score = 2, give only diazepam. For agitation assessment score 3 or greater, give with haloperidol.[a] Oral preferred. Give IM if oral not feasible.
	Ziprasidone	20 mg	PO	For agitation assessment score 3 or greater, give with lorazepam.[c] Oral preferred. Give IM if oral not feasible.
		10 mg	IM[d]	

Abbreviations: IM, intramuscular; PO, appropriate oral.
[a]To be given with ziprasidone instead of haloperidol if ordered.
[b]Utilized only if patient has allergies to (or has failed to respond to) haloperidol or lorazepam.
[c]To be given with diazepam instead of lorazepam if ordered.
[d]Maximum dose 40 mg per 24 hours.

spent in physical restraints changed from 11.3 to 11.97, and neither *P* value was statistically significant (95% CI = [–7.28 to 60.31]; $P = 0.124$, and 95% CI = [–19.4 to 18.1]; $P = 0.994$, respectively).

Stratification Based on Illness, Scheduled Medication Use, and Minutes in Seclusions or Physical Restraints

Because of the lower numbers of other mental illnesses seen in our population, we only adjusted for those with a schizophrenia spectrum disorder (the single largest number of patients included in the analyses). This post hoc analysis showed that the *P* value of mean PRN administrations pre-BAM versus BAM reached statistical significance (9.31 ± 13.1 vs 5.48 ± 10.1; 95% CI = [0.324-7.33]; $P = 0.032$), while the median PRN administrations trended toward significance ($P = 0.055$).

When stratified based on scheduled medication, it appeared that those on scheduled benzodiazepines required more PRNs; 4.1% of those not on scheduled benzodiazepines required 11 to 20 PRNs, compared with 24.6% of those on scheduled benzodiazepines, whereas 29.3% of those not on scheduled benzodiazepines required zero PRNs versus only 9.8% on scheduled benzodiazepines. The inverse appeared to be true for scheduled antipsychotics, however; those on scheduled antipsychotics appeared to require fewer PRNs than those not on scheduled antipsychotics. No association was noted between PRN use and scheduled mood stabilizers, preprotocol or postprotocol implementation (Table 4).

The number of minutes spent in seclusion or physical restraints aligned highly with the number of PRNs administered. Patients not receiving any PRNs during pre-BAM or BAM had a far lower incidence of requiring physical restraints or seclusion (0 pre-BAM patients [of 42 patients], 1 BAM patient (of 77 patients)—total of 60 minutes spent in seclusion). For patients receiving between 1 and 10 PRNs, the average duration spent in physical restraints was 5.4 minutes, and the average duration spent in seclusion was 41.3 minutes. For those receiving >10 PRNs, the average duration spent in physical restraints was 45.4 minutes, and the average duration spent in seclusion was 178.1 minutes. No significant correlations were noted between specific diagnoses and the risk of physical restraints or seclusion.

Discussion

The application of the standardized BAM protocol appeared to be effective in reducing the total number of PRNs among the average high-acuity adult psychiatric inpatients. This is clinically relevant as PRN medications come with excess sedation, undue extrapyramidal adverse effects, and increased health care expenses. As the rates of seclusion and physical restraints between pre-BAM and BAM groups were similar, giving fewer PRNs may be as effective as giving more PRNs, while reducing the risk for adverse effects. Despite mixed evidence supporting the use of PRN medications for agitation, their prescribing remains commonplace.[5,36,37] The majority of the patients included in the study required at least 1 PRN for agitation during their stay (77.2% pre-BAM and 60% BAM). This is not unlike the

Table 3. Demographic Data Pre-BAM and BAM Implementation.

Factor	Pre-BAM (n = 184)	BAM (n = 193)
Age, years	35.5 (11.2)	35 (10.7)
Sex, male	121 (65.8%)	139 (72%)
Race		
AA	127 (69%)	141 (73.1%)
Caucasian	55 (29.9%)	50 (25.9%)
Other	3 (1.6%)	2 (1%)
Ethnicity		
Hispanic	1 (0.5%)	0 (0%)
Non-Hispanic	183 (99.5%)	193 (100%)
LOS, days	16.5 (13.4)	15.7 (13.4)
Discharged to state hospital	24 (13%)	16 (8.3%)
Diagnosis		
Schizophrenia/schizoaffective disorder/schizophreniform	81 (44%)	91 (47.2%)
Psychosis NOS/brief psychotic disorder/delusional disorder	8 (4.4%)	32 (16.6%)
SIPD/SIMD	42 (22.8%)	49 (25.4%)
Bipolar disorder	42 (22.8%)	22 (11.4%)
MDD/PTSD/OCD	15 (8.2%)	19 (9.8%)
ID/cognitive disorder/impulse control disorder	16 (8.7%)	26 (13.5%)
Personality disorder	9 (4.9%)	7 (3.6%)
Scheduled benzodiazepine	62 (33.7%)	50 (26%)
Scheduled antipsychotic	167 (90.8%)	176 (91.2%)
Scheduled mood stabilizer	65 (35.3%)	65 (33.7%)

Data are mean (SD) or n (%).
Abbreviations: AA, African American; BAM, Birmingham Agitation Management; ID, intellectual disability; LOS, length of stay; MDD, major depressive disorder; NOS, not otherwise specified; OCD, obsessive compulsive disorder; PTSD, posttraumatic stress disorder; SIMD, substance-induced mood disorder; SIPD, substance-induced psychotic disorder.

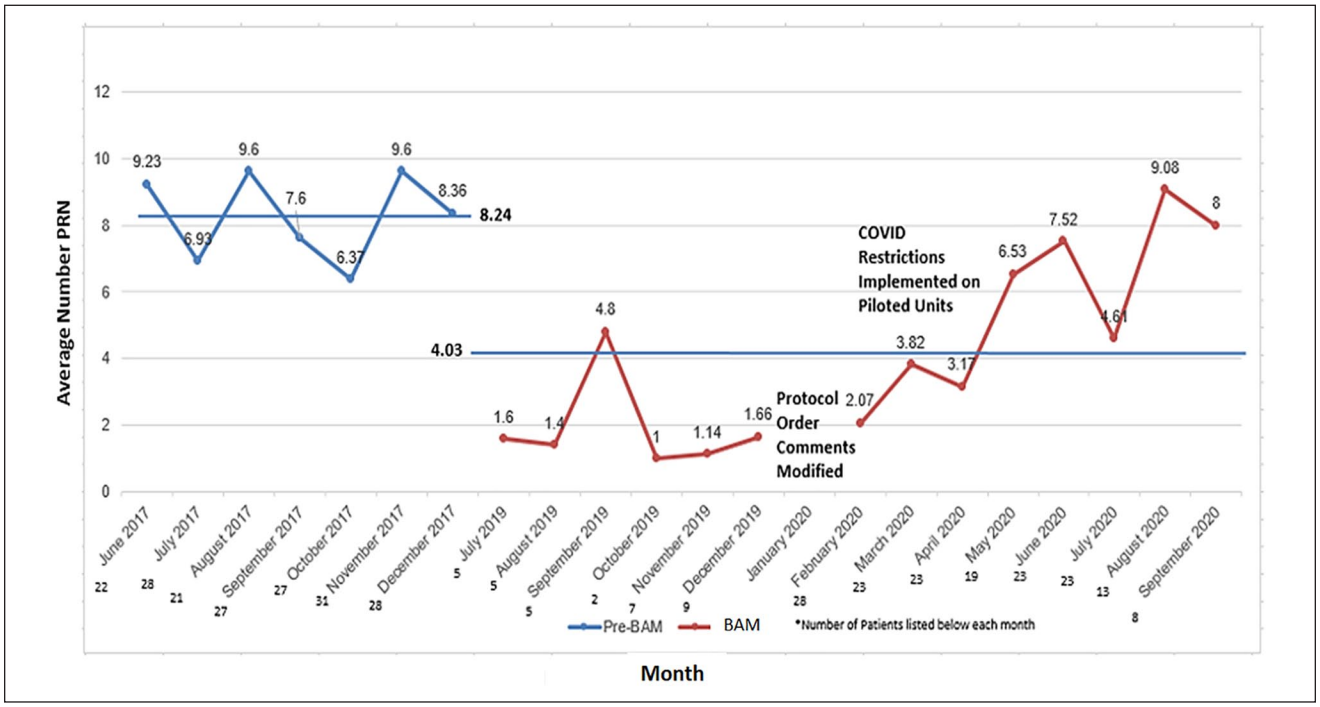

Figure 2. Average number of PRNs administered per month pre-BAM and during BAM.
Abbreviations: BAM, Birmingham Agitation Management; PRN, pro re nata.

Table 4. Primary, Secondary, and Post Hoc Data Analyses.

	Mean	SD	Comparison of means (2-sided *t* test) *P* value	95% CI	Comparison of medians (Mann–Whitney *U*) *P* value
Primary analysis (All PRNs)					
Pre-BAM	8.21	11.3	<0.001	[1.68-5.75]	<0.001
BAM	4.49	8.7			
Secondary analysis (select PRNs)					
Pre-BAM	5.57	7	0.190	[–0.534 to 2.68]	0.023
BAM	4.49	8.7			
Secondary analysis (minutes spent in seclusion)					
Pre-BAM	69.23	208.8	0.124	[–7.28 to 60.31]	0.393
BAM	42.72	113.06			
Secondary analysis (minutes spent in physical restraints)					
Pre-BAM	11.30	82.9	0.994	[–19.4 to 18.1]	0.773
BAM	11.97	100.5			
Post hoc analysis (PRN administration schizophrenia spectrum disorder only)					
Pre-BAM	9.31	13.1	0.032	0.324-7.33	0.055
BAM	5.48	10.1			

Abbreviations: BAM, Birmingham Agitation Management; CI, confidence interval; PRN, pro re nata.

percentage seen in a 2008 study, where 83.8% of acute psychiatric inpatients required at least 1 PRN administration.[38] While this range has been noted to vary within some other studies, we believe that the high acuity and large number of patients with a schizophrenia spectrum disorder assessed within our study may have influenced the percentage.[20]

When comparing only those PRN medications available for use during the BAM protocol to pre-BAM, the *P* value of the average PRN administration did not reach statistical significance, but possibly trended toward significance. The lack of a specific medication-related effect might indicate that the standardization of PRN medication administration is more relevant than the specific medication that is given. More data are needed to determine the specific effect of each medication.

In general, aside from those with a schizophrenia spectrum disorder, it did not appear that any diagnosis was strongly associated with the number of PRNs administered preimplementation or postimplementation. This is based on the fact that the percentage of BAM patients requiring 0, 1-10, 11-20, or >20 PRNs was relatively comparable, regardless of whether or not they had that specific diagnosis. Overall, patients may have had multiple diagnoses, making this interpretation difficult. Furthermore, our stratification of diagnoses may have altered the percentages. The *Diagnostic and Statistical Manual of Mental Disorders*, Fifth Edition (*DSM*-5) categorizes brief psychotic disorder, psychosis NOS, and delusional disorder as part of schizophrenia spectrum disorders; however, for the sake of this analysis, we separated out these diagnoses from others in the category. At this time, more investigation is needed to determine whether a specific diagnosis was affected by PRN administration standardization as analysis was limited by the small number of patients in certain diagnosis groups.

A larger number of PRN lorazepam 2 mg PO, 3 mg PO, and 3 mg IM was given in comparison with lorazepam 2 mg IM, while a larger number of haloperidol 5 mg IM was given in comparison with 5 mg PO. While administered far less frequently, these correlates were also seen with diazepam and ziprasidone. This likely indicates that at lower BVC scores, patients were more receptive to taking PO benzodiazepines than at higher BVC scores where patients were forced to receive IM benzodiazepines plus an antipsychotic. In terms of scheduled medications, while the number of PRNs given to those patients on scheduled benzodiazepines was higher than those not on scheduled benzodiazepines, this number was comparable between preimplementation and postimplementation, likely implying that those patients requiring scheduled benzodiazepines were also more likely to require PRNs. Thus, scheduled benzodiazepines could have been a marker for a generally higher baseline agitation level. Scheduled antipsychotic use may have potentially prevented patients from receiving 11 or more PRNs during their admission, conceivably due to antipsychotics more directly treating the underlying psychiatric condition.

Our project did not demonstrate the hypothesized reduction in the rate or minutes spent in seclusion or physical restraints. This may be related to the reactive, rather than proactive nature of PRN medication administrations. The literature continues to be very mixed on whether or not PRN medication controls the patient who is escalating toward full loss of behavioral control.[5,36,37] In addition, it was evident that many of the analyzed patients

were treatment-resistant, based on the length of stay and commitment requirements. This might have led to inherently higher rates of seclusion. We also cannot fully deduce the impact of COVID-19 and new measures that were required pre-BAM versus BAM on our results.

Although our BAM protocol appears to be the first time the BVC has been utilized to correlate agitation severity with medication administration, it is not the first time a scale has been implemented for administration criterion. A 2008 study utilized the PANSS-EC as a guide for PRN administration.[39] While this study did not find a difference in the mean number of PRNs administered preimplementation and postimplementation, it was limited by the small number of patients included in each group (35 preimplementation vs 41 postimplementation), and did not standardize the regimen of medications utilized. Our study is also not the first to assess the impact of intervention on PRN medication administration. In a study examining the impact of guidance for IM administration of agitation medications, the authors found that strict oversight decreased the total number of PRN agitation administrations from a monthly average of 201 to 135 per 1000 days of care ($P = 0.002$).[40] Furthermore, IM administrations decreased by half, whereas oral administrations decreased by 30% ($P = 0.002$ for both).[40] Separately, a 2003 study determined that when PRN medications were not allowed, and only "now" orders were permitted, the total number of unscheduled agitation medications decreased from 1812 to 976 (adjusted mean doses per admission decreased from 7.8 to 4.3).[41] As with our project, these authors did not find a statistically significant difference in the assault or restraint rates prepolicy and postpolicy implementation.[40,41] Finally, in a double-blind, randomized study of lorazepam alone versus lorazepam plus haloperidol, the authors utilized the Overt Aggression Scale at baseline to score the patient prior to medication administration and 30, 60, 120, and 180 minutes after.[34] While only 20 patients were included, the combination was noted to be more effective than the use of lorazepam itself ($P < 0.05$).[34]

Not dissimilar from our BAM protocol is the CIWA-Ar, which may be tied to the administration of a benzodiazepine. Similar to the CIWA-Ar, the BVC can help with objective agitation assessment by categorically placing a numerical value to agitation severity. This can help to remove the subjectivity of assessment, which may be a large reason for differences in agitation treatment. Studies evaluating the administration of scheduled versus symptom-triggered benzodiazepines for alcohol withdrawal demonstrated either equivalent or superior clinical end points with PRN usage, while utilizing fewer doses.[42,43] Effectiveness of both the BAM and CIWA-Ar protocols may be limited, however, by nursing training and acceptance.[44-46] In addition, subjectivity of the scaled items may still present a problem (ie, defining "boisterousness" on the BVC, or "moderate" versus "severe" tactile disturbances on the CIWA-Ar).[47] Frequent input from the treatment team with judicious use of these protocols may enhance their overall effectiveness. Providing timely feedback of relevant clinical data was shown to be effective in reducing PRN psychotropic medications.[48] As previously noted, 51 patients were excluded for issues with incorrect doses or medications being administered after protocol implementation. Root cause analysis of these violations led us to modify our order sentences in a way that clarified when each specific dose or medication should be given, based on the BVC score, rectifying this issue (Figure 2).

Key limitations to our study were due to the retrospective nature of the comparison group. We were unable to fully assess the effectiveness of the PRN medications given, or the impact of verbal de-escalation, on agitation. Because the comparison groups had different participants, we are unable to conclusively determine the effectiveness of the overall protocol; but the close similarities between groups likely imply benefit from the protocol. A transition in software led to an inability to determine assault data specific to the included patients, precluding us from determining the effect of the BAM protocol on this measure. Finally, we did not evaluate the rates of tolerance, extrapyramidal adverse effects, or oversedation pre-BAM for comparison. However, we do not believe the rates of dystonic reactions were higher in our study after BAM implementation, despite the absence of concomitant anticholinergic medication administration. Our rationale here is that benzodiazepines have been shown to mitigate extrapyramidal symptom (EPS), as seen within the Battaglia et al[33] study, which found the rates of EPS with haloperidol alone were 20%, versus 6% when administered with lorazepam.[14,49]

For future directions of this protocol, we plan to evaluate ziprasidone plus benzodiazepine versus haloperidol plus benzodiazepine efficacy for medication-related effects by reassessing the BVC score shortly after medication administration. With more patients, we hope to assess specific diagnoses for BAM benefits or ineffectiveness. We will also attempt to utilize this protocol for our geriatric and adolescent patients, likely with altered medicated regimens in reduced doses. In addition, we plan to publish the results of our nursing, staff, and patient satisfaction surveys after the implementation BAM protocol. Preliminary results of these surveys have been very positive.

Conclusion and Relevance

In this piloted project, we found that the BAM protocol was effective in decreasing the total number of PRNs administered for agitation. The standardization of administration may be more important than the specific medications utilized. Whereas the rates of seclusion and physical restraints did not decrease, they also did not increase, likely denoting

that those fewer medications could be as effective as more medications. While further studies are required to validate these findings, clinicians may look to implement the BAM or similar protocols in their acute inpatient practices to improve patient care.

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References

1. Swanson JW, Holzer CE 3rd, Ganju VK, Jono RT. Violence and psychiatric disorder in the community: evidence from the Epidemiologic Catchment Area surveys. *Hosp Community Psychiatry*. 1990;41(7):761-770. doi:10.1176/ps.41.7.761.
2. Tardiff K, Sweillam A. Assaultive behavior among chronic inpatients. *Am J Psychiatry*. 1982;139(2):212-215. doi:10.1176/ajp.139.2.212.
3. Zeller SL, Rhoades RW. Systematic reviews of assessment measures and pharmacologic treatments for agitation. *Clin Ther*. 2010;32(3):403-425. doi:10.1016/j.clinthera.2010.03.006.
4. Roppolo LP, Morris DW, Khan F, et al. Improving the management of acutely agitated patients in the emergency department through implementation of Project BETA (Best Practices in the Evaluation and Treatment of Agitation). *J Am Coll Emerg Physicians Open*. 2020;1(5):898-907. doi:10.1002/emp2.12138.
5. Srivastava A. Limited evidence for the effectiveness of p.r.n. Medications among psychiatric inpatients. *J Psychiatr Pract*. 2009;15(3):193-201. doi:10.1097/01.pra.0000351879.52883.10.
6. Patel MX, Sethi FN, Barnes TR, et al. Joint BAP NAPICU evidence-based consensus guidelines for the clinical management of acute disturbance: de-escalation and rapid tranquillisation. *J Psychopharmacol*. 2018;32(6):601-640. doi:10.1177/0269881118776738.
7. Almvik R, Woods P, Rasmussen K. Assessing risk for imminent violence in the elderly: the Brøset Violence Checklist. *Int J Geriatr Psychiatry*. 2007;22(9):862-867. doi:10.1002/gps.1753.
8. Sarver WL, Radziewicz R, Coyne G, Colon K, Mantz L. Implementation of the Brøset Violence Checklist on an acute psychiatric unit. *J Am Psychiatr Nurses Assoc*. 2019;25(6):476-486. doi:10.1177/1078390318820668.
9. Allen MH. Managing the agitated psychotic patient: a reappraisal of the evidence. *J Clin Psychiatry*. 2000;61(suppl 14):11-20.
10. Lukens TW, Wolf SJ, Edlow JA, et al. Clinical policy: critical issues in the diagnosis and management of the adult psychiatric patient in the emergency department. *Ann Emerg Med*. 2006;47(1):79-99. doi:10.1016/j.annemergmed.2005.10.002.
11. Allen MH, Currier GW, Carpenter D, Ross RW, Docherty JP. The expert consensus guideline series. Treatment of behavioral emergencies 2005. *J Psychiatr Pract*. 2005;11(suppl 1):5-108. doi:10.1097/00131746-200511001-00002.
12. Gautam M, Patel S, Sablaban I. Intramuscular B52. *Prim Care Companion CNS Disord*. 2021;23(3):20br02855. doi:10.4088/PCC.20br02855.
13. Battaglia J. Pharmacological management of acute agitation. *Drugs*. 2005;65(9):1207-1222. doi:10.2165/00003495-200565090-00003.
14. Wilson MP, Pepper D, Currier GW, Holloman GH Jr, Feifel D. The psychopharmacology of agitation: consensus statement of the American Association for Emergency Psychiatry Project Beta Psychopharmacology Workgroup. *West J Emerg Med*. 2012;13(1):26-34. doi:10.5811/westjem.2011.9.6866.
15. Peisah C, Chan DK, McKay R, Kurrle SE, Reutens SG. Practical guidelines for the acute emergency sedation of the severely agitated older patient. *Intern Med J*. 2011;41(9):651-657. doi:10.1111/j.1445-5994.2011.02560.x.
16. Ostinelli EG, Brooke-Powney MJ, Li X, Adams CE. Haloperidol for psychosis-induced aggression or agitation (rapid tranquillisation). *Cochrane Database Syst Rev*. 2017;7(7):Cd009377. doi:10.1002/14651858.CD009377.pub3.
17. State Operations Manual. Survey Protocol, Regulations and Interpretive Guidelines for Hospitals. State Operations Manual; 2020:1-562. Accessed July 26, 2022. https://www.cms.gov/Regulations-and-Guidance/Guidance/Manuals/downloads/som107ap_a_hospitals.pdf.
18. The Joint Commission Division of Healthcare Improvement. Quick Safety: de-escalation in health care. 2019;(47):1-4. Accessed July 26, 2022. https://www.jointcommission.org/resources/news-and-multimedia/newsletters/newsletters/quick-safety/quick-safety-47-deescalation-in-health-care/.
19. Hilton MF, Whiteford HA. Pro re nata medication for psychiatric inpatients: time to act. *Aust N Z J Psychiatry*. 2008;42(7):555-564. doi:10.1080/00048670802119804.
20. Baker JA, Lovell K, Harris N. A best-evidence synthesis review of the administration of psychotropic pro re nata (PRN) medication in in-patient mental health settings. *J Clin Nurs*. 2008;17(9):1122-1131. doi:10.1111/j.1365-2702.2007.02236.x.
21. Geffen J, Sorensen L, Stokes J, Cameron A, Roberts MS, Geffen L. Pro re nata medication for psychoses: an audit of practice in two metropolitan hospitals. *Aust N Z J Psychiatry*. 2002;36(5):649-656. doi:10.1046/j.1440-1614.2002.01069.x.
22. Curtis J, Capp K. Administration of "as needed" psychotropic medication: a retrospective study. *Int J Ment Health Nurs*. 2003;12(3):229-234. doi:10.1046/j.1440-0979.2003.00293.x.
23. Geffen J, Cameron A, Sorensen L, Stokes J, Roberts MS, Geffen L. Pro re nata medication for psychoses: the

knowledge and beliefs of doctors and nurses. *Aust N Z J Psychiatry*. 2002;36(5):642-648. doi:10.1046/j.1440-1614.2002.01068.x.
24. Martin K, Arora V, Fischler I, Tremblay R. Analysis of non-pharmacological interventions attempted prior to pro re nata medication use. *Int J Ment Health Nurs*. 2018;27(1):296-302. doi:10.1111/inm.12320.
25. Casol M, Tong A, Ng JCY, McGloin R. Characterization of psychotropic PRN medications in a Canadian psychiatric intensive care unit. *J Am Psychiatr Nurses Assoc*. 2021:1078390321994668. doi:10.1177/1078390321994668.
26. Fujita J, Nishida A, Sakata M, Noda T, Ito H. Excessive dosing and polypharmacy of antipsychotics caused by pro re nata in agitated patients with schizophrenia. *Psychiatry Clin Neurosci*. 2013;67(5):345-351. doi:10.1111/pcn.12056.
27. Alexander J, Tharyan P, Adams C, John T, Mol C, Philip J. Rapid tranquillisation of violent or agitated patients in a psychiatric emergency setting. Pragmatic randomised trial of intramuscular lorazepam v. haloperidol plus promethazine. *Br J Psychiatry*. 2004;185:63-69. doi:10.1192/bjp.185.1.63.
28. McGreevey JD 3rd. Order sets in electronic health records: principles of good practice. *Chest*. 2013;143(1):228-235. doi:10.1378/chest.12-0949.
29. Hvidhjelm J, Sestoft D, Skovgaard LT, Bue Bjorner J. Sensitivity and specificity of the Brøset Violence Checklist as predictor of violence in forensic psychiatry. *Nord J Psychiatry*. 2014;68(8):536-542. doi:10.3109/08039488.2014.880942.
30. Partridge B, Affleck J. Predicting aggressive patient behaviour in a hospital emergency department: an empirical study of security officers using the Brøset Violence Checklist. *Australas Emerg Care*. 2018;21(1):31-35. doi:10.1016/j.auec.2017.11.001.
31. Yao X, Li Z, Arthur D, Hu L, An FR, Cheng G. Acceptability and psychometric properties of Brøset Violence Checklist in psychiatric care settings in China. *J Psychiatr Ment Health Nurs*. 2014;21(9):848-855. doi:10.1111/jpm.12132.
32. Marder SR, Sorsaburu S, Dunayevich E, et al. Case reports of postmarketing adverse event experiences with olanzapine intramuscular treatment in patients with agitation. *J Clin Psychiatry*. 2010;71(4):433-441. doi:10.4088/JCP.08m04411gry.
33. Battaglia J, Moss S, Rush J, et al. Haloperidol, lorazepam, or both for psychotic agitation? A multicenter, prospective, double-blind, emergency department study. *Am J Emerg Med*. 1997;15(4):335-340. doi:10.1016/s0735-6757(97)90119-4.
34. Bieniek SA, Ownby RL, Penalver A, Dominguez RA. A double-blind study of lorazepam versus the combination of haloperidol and lorazepam in managing agitation. *Pharmacotherapy*. 1998;18(1):57-62.
35. Meehan K, Zhang F, David S, et al. A double-blind, randomized comparison of the efficacy and safety of intramuscular injections of olanzapine, lorazepam, or placebo in treating acutely agitated patients diagnosed with bipolar mania. *J Clin Psychopharmacol*. 2001;21(4):389-397. doi:10.1097/00004714-200108000-00006.
36. Remick RA. Treatment resistant depression. *Psychiatr J Univ Ott*. 1989;14(2):394-396.
37. Martin K, Arora V, Fischler I, Tremblay R. Descriptive analysis of pro re nata medication use at a Canadian psychiatric hospital. *Int J Ment Health Nurs*. 2017;26(4):402-408. doi:10.1111/inm.12265.
38. Stein-Parbury J, Reid K, Smith N, Mouhanna D, Lamont F. Use of pro re nata medications in acute inpatient care. *Aust N Z J Psychiatry*. 2008;42(4):283-292. doi:10.1080/00048670701881553.
39. Chaichan W. Evaluation of the use of the positive and negative syndrome scale-excited component as a criterion for administration of p.r.n. medication. *J Psychiatr Pract*. 2008;14(2):105-113. doi:10.1097/01.pra.0000314317.79544.b4.
40. Hayes A, Russ MJ. Discontinuing the use of PRN intramuscular medication for agitation in an acute psychiatric hospital. *Psychiatr Q*. 2016;87(1):25-29. doi:10.1007/s11126-015-9359-1.
41. Thapa PB, Palmer SL, Owen RR, Huntley AL, Clardy JA, Miller LH. P.R.N. (As-needed) orders and exposure of psychiatric inpatients to unnecessary psychotropic medications. *Psychiatr Serv*. 2003;54(9):1282-1286. doi:10.1176/appi.ps.54.9.1282.
42. Sen S, Grgurich P, Tulolo A, et al. A symptom-triggered benzodiazepine protocol utilizing SAS and CIWA-Ar scoring for the treatment of alcohol withdrawal syndrome in the critically ill. *Ann Pharmacother*. 2017;51(2):101-110. doi:10.1177/1060028016672036.
43. Daeppen JB, Gache P, Landry U, et al. Symptom-triggered vs fixed-schedule doses of benzodiazepine for alcohol withdrawal: a randomized treatment trial. *Arch Intern Med*. 2002;162(10):1117-1121. doi:10.1001/archinte.162.10.1117.
44. Bacon O, Robert S, VandenBerg A. Evaluating nursing satisfaction and utilization of the Clinical Institute Withdrawal Assessment for Alcohol, revised version (CIWA-Ar). *Ment Health Clin*. 2016;6(3):114-119. doi:10.9740/mhc.2016.05.114.
45. Thompson C, McLeod SL, Perelman V, et al. Evaluation of a web-based educational program to teach the identification and management of alcohol withdrawal in the emergency department. *AEM Educ Train*. 2019;3(1):50-57. doi:10.1002/aet2.10202.
46. Daly M, Kermode S, Reilly D. Evaluation of clinical practice improvement programs for nurses for the management of alcohol withdrawal in hospitals. *Contemp Nurse*. 2009;31(2):98-107. doi:10.5172/conu.673.31.2.98.
47. Eloma AS, Tucciarone JM, Hayes EM, Bronson BD. Evaluation of the appropriate use of a CIWA-Ar alcohol withdrawal protocol in the general hospital setting. *Am J Drug Alcohol Abuse*. 2018;44(4):418-425. doi:10.1080/00952990.2017.1362418.
48. Friedman R, Nurenberg JR, Birnbaum S, Schleifer SJ. Using structured clinical feedback to encourage alternatives to use of "P.R.N." medication in a state psychiatric hospital. *J Psychiatr Pract*. 2012;18(5):381-387. doi:10.1097/01.pra.0000419823.69914.c7.
49. Holloman LC, Marder SR. Management of acute extrapyramidal effects induced by antipsychotic drugs. *Am J Health Syst Pharm*. 1997;54(21):2461-2477. doi:10.1093/ajhp/54.21.2461.